

Question on Merge Sorting.

Leetcode: Count of Smaller Numbers after self.

Count of Smaller numbers than self.

For $O(n \log n)$ $\rightarrow$ Fenwick / Segment Tree

$\rightarrow$ Binary Search

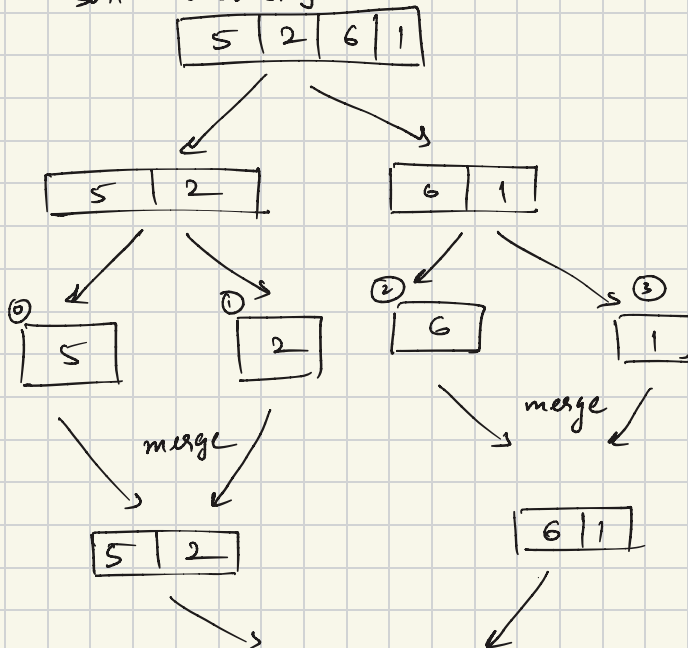
$\rightarrow$ Sorting

array containing elements right to it.

Input: [5, 2, 6, 1]

output: [2, 1, 1, 0]

5 has 2 and 1 on the right
sort in decreasing order.



count ++
happens when
left ptr < Right ptr.

while merging

5	2
---	---

 so 2 is on the right of 5 so we do $\text{count}[\text{index of } 5]++$
similarly for

6	1
---	---

 $\rightarrow \text{count}[\text{index of } 6]++$

while merging

5	2
---	---

6	1
---	---

6	.	.	.
---	---	---	---

 $\uparrow$ left $\uparrow$ right.

we do $\text{count}++$ when $\text{left} > \text{right}$

5	2
---	---

6	1
---	---

 $\rightarrow$

6	5	.	.
---	---	---	---

 $\uparrow$ left $\uparrow$ right.

now. $\text{left} > \text{right}$ i.e. 1 is smaller than 5 and on the right of 5.
 $\text{count}[\text{index of } 5]++$

5	2
---	---

6	1
---	---

 $\rightarrow$

6	5	2	.
---	---	---	---

 $\uparrow$ left $\uparrow$ right

$\text{left} > \text{right}$ -
so. $\text{count}[\text{index of } 2]++$

simply use `vector<pair<int, int>> numPair`
`<nums[i], i>`